Summary and Methodology

To begin with, it is important to clarify that there was no direct access to comprehensive data on injuries or accidents, and the available information was highly dispersed—mostly small news articles reporting one or two incidents across 15 years for about 60 National Parks. Therefore, the analysis focused on **wildlife encounters** as the most consistent and comparable indicator of risk.

The datasets were obtained from the **National Parks official website** and **Open Canada**. After cleaning and standardizing, we identified **57 National Parks** that appeared in both sources and contained sufficient data to be analyzed. These were used to determine which parks presented higher relative risk based on historical records.

Visitors and wildlife encounter datasets were merged. An extensive search was also conducted across **CHIRPP**, **CIHI**, and other injury registries, but none specified whether incidents occurred inside National Parks. Similarly, no individual park provides open access to detailed records of accidents or injuries.

In some cases, data on mountain incidents were found through National Parks reports. Missing visitor counts were filled by interpolation — using the mean between two non-consecutive years or, when only one side was available, replicating the nearest known value. Park names were standardized to ensure consistency when merging the datasets.

Finally, a **risk score** was calculated as:

$$(\text{number of incidents}) \times 100000/(\text{number of visitors})$$

The resulting scores were compiled in an Excel file. The **average value across available years** for each park was used to represent its relative danger level, based on all the information publicly accessible online.

Key Findings

- Based on historical data, **Wapusk National Park** appears to be the most dangerous overall.
- Most parks are relatively safe — for example, **Rideau Canal National Historic Site** and **Chambly Canal National Historic Site** reported **zero incidents over the past fifteen years**.
- **Banff National Park of Canada** recorded the highest number of **mountain-related accidents** among all sites.
- **Additional note:** Most wildlife incidents were associated with **black bears**.

Recommendations for Future Study

It is important to acknowledge the limitations of the currently available data. Moving forward, the main challenge is not simply to obtain existing information but to **build a more complete dataset** using data extracted from **news reports and public records**. This approach would be time-consuming and technically demanding, yet it represents the **most feasible method** to capture incident data that is not publicly available through official channels.

Databases:

https://open.canada.ca/data/en/dataset/743a0b4a-9e33-4b12-981a-9f9fd3dd1680/resource/eaadb900-e1e5-4844-9617-4b8d76f0ea80 (wildlife incidents, main database)

https://parks.canada.ca/pn-np/mtn/securiteenmontagne-mountainsafety/accidents (all the incidents considered in this work in mountains)

https://open.canada.ca/data/en/dataset/eaa1ff9c-a07c-4292-911c-612f39d81afb (number of visitors, five datasets)

https://parks.canada.ca/ (general information)

Note: Many more sources were considered, such as CIHI and CHIRPP websites, but here I only mention those were information feasible to relate it with National Park accidents.